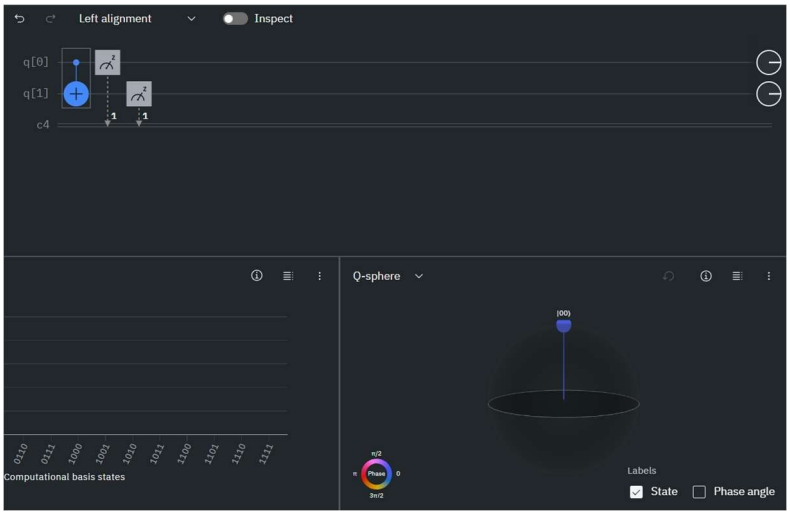
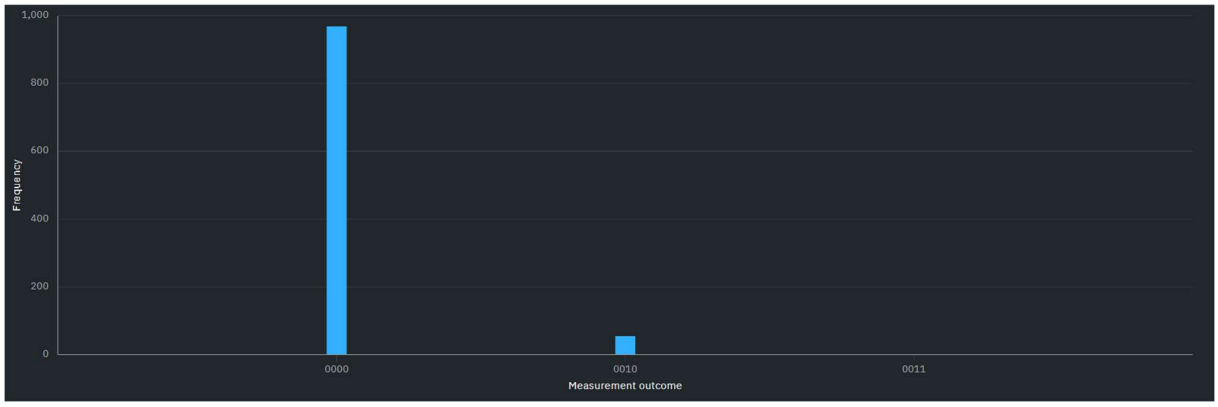
QC Assignment 4

Experiment 1 :

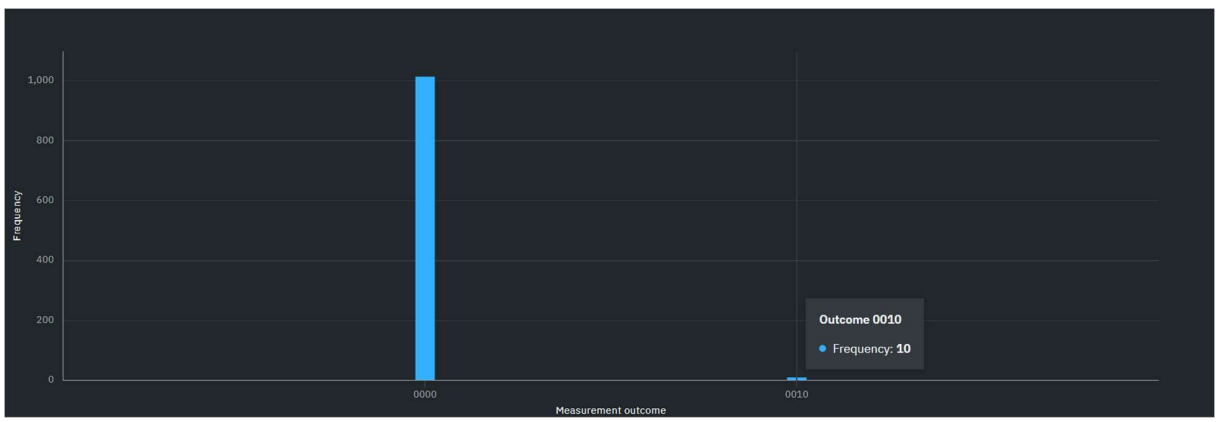
CNOT:



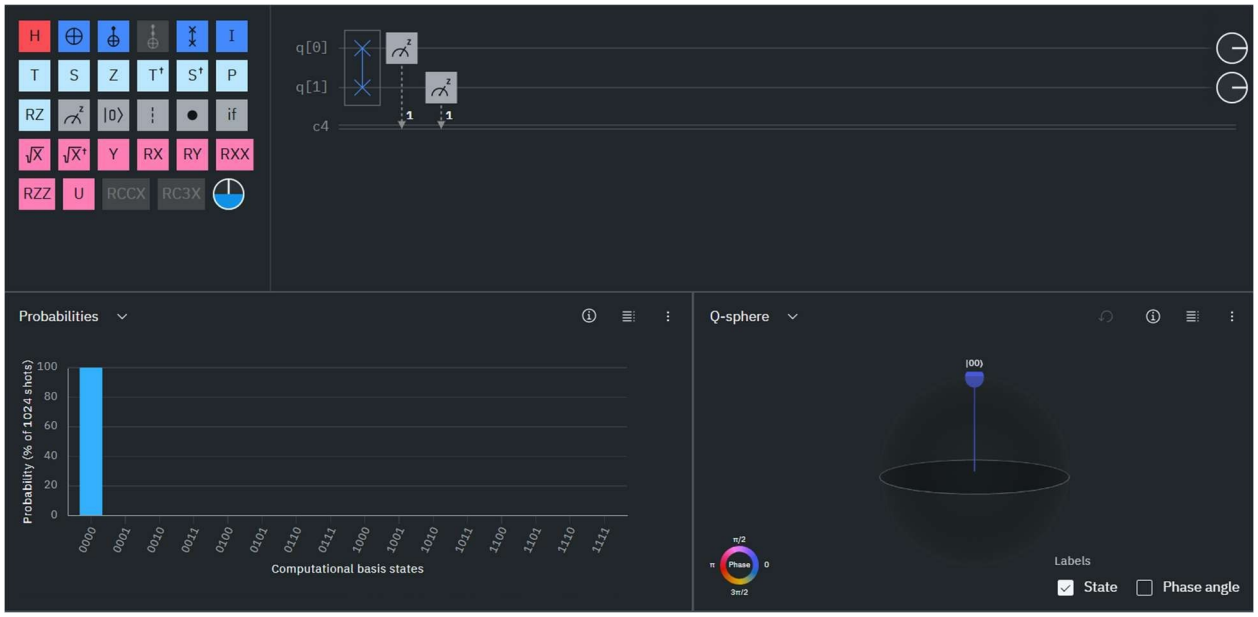


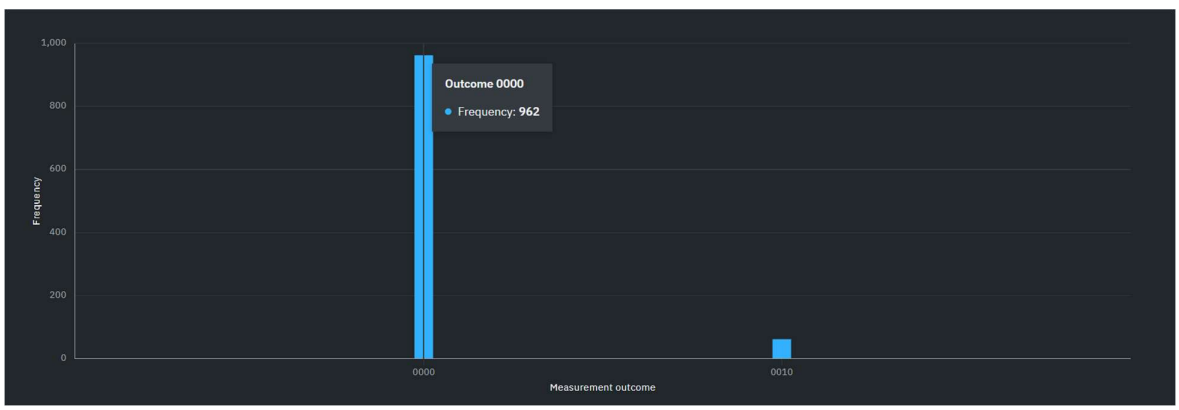
CZ:

q[0]
q[1]
c4
H
H
1
1
Q-sphere
|00⟩
0110
0111
1000
1001
1010
1011
1100
1101
1110
1111
omputational basis states
π/2
π
Phase
0
3π/2
Labels
State
Phase angle

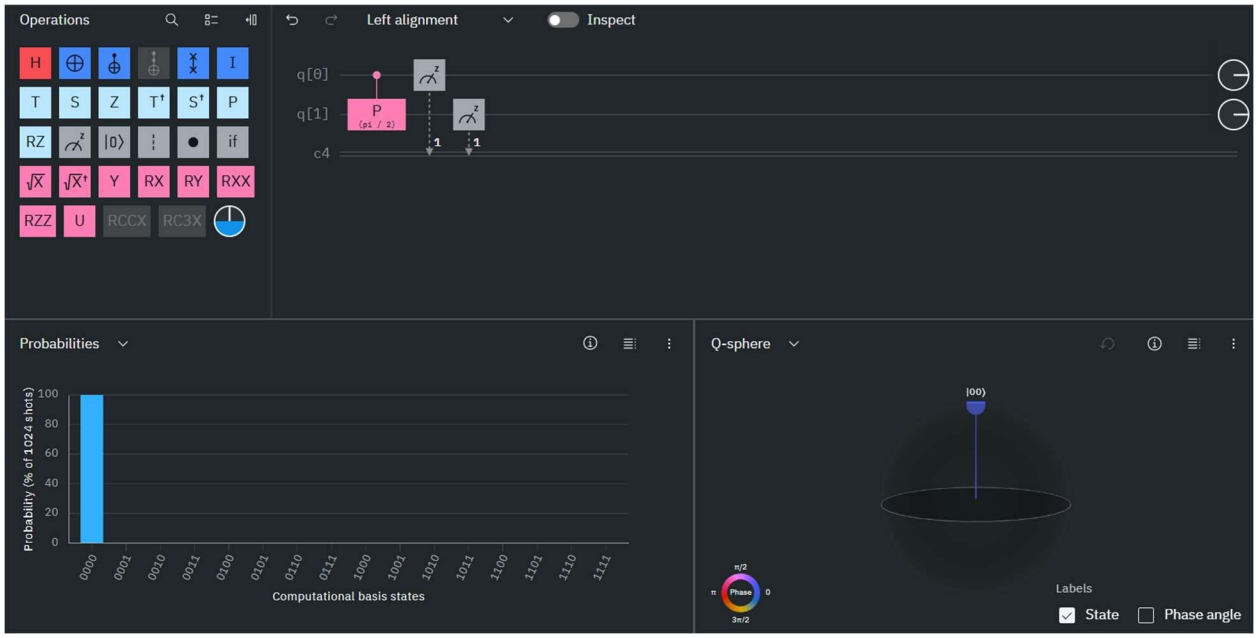

1,000
800
600
400
200
0
Frequency
0000
0010
Measurement outcome
Outcome 0010
Frequency: 10

SWAP:


H
I
T
S
Z
T†
S†
P
RZ
|0⟩
if
√X
√X†
Y
RX
RY
RXX
RZZ
U
RCCX
RC3X
q[0]
q[1]
c4
1
1
Probabilities
Probability (% of 1024 shots)
100
80
60
40
20
0
0000
0001
0010
0011
0100
0101
0110
0111
1000
1001
1010
1011
1100
1101
1110
1111
Computational basis states
Q-sphere
|00⟩
π/2
π
Phase
0
3π/2
Labels
State
Phase angle

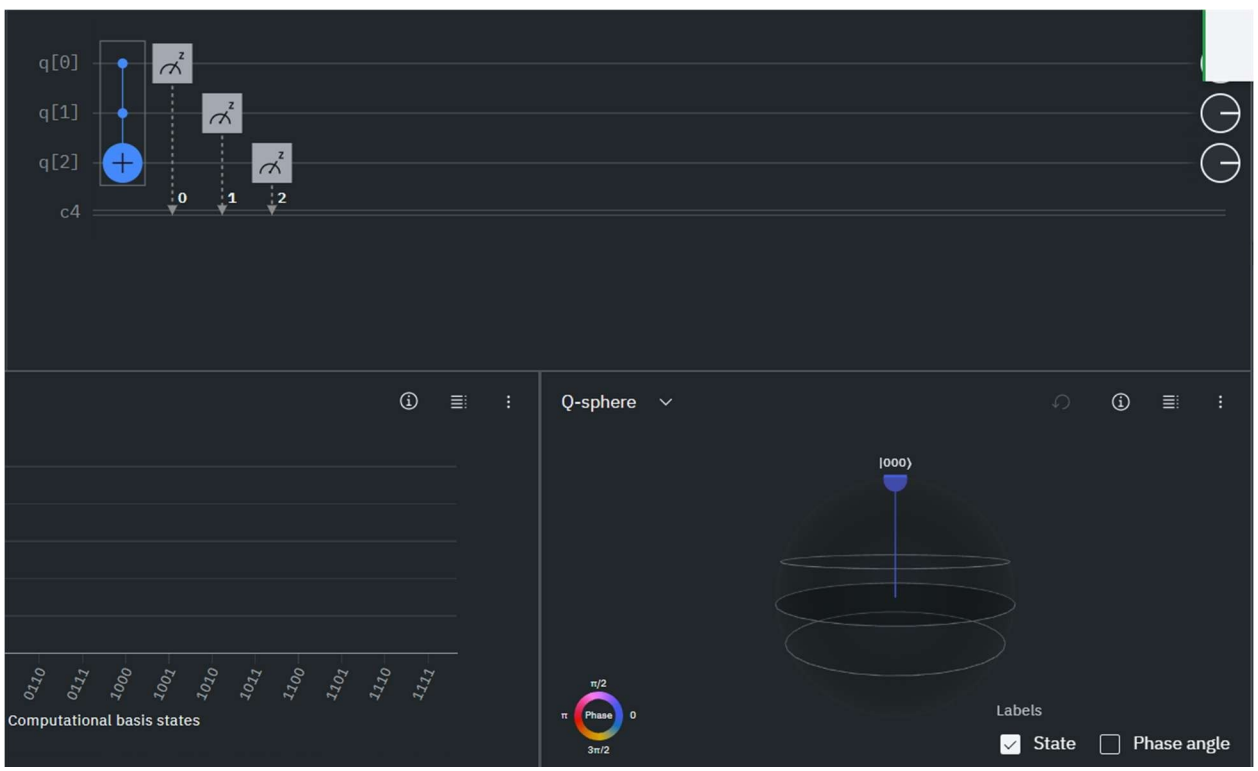

1,000
800
600
400
200
0
Frequency
Outcome 0000
Frequency: 962
0000
0010
Measurement outcome

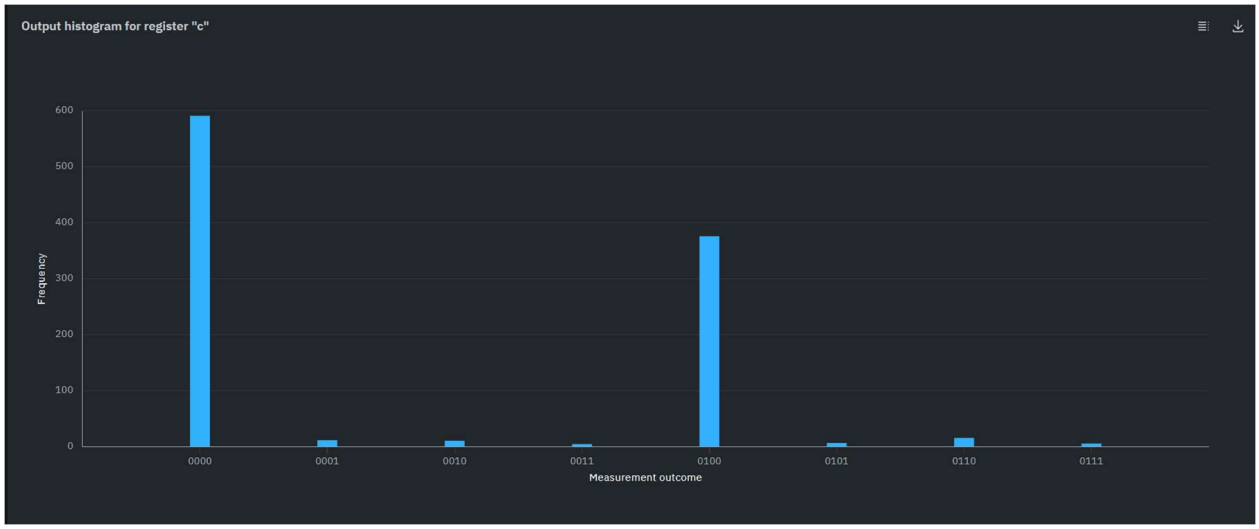
CPHASE:



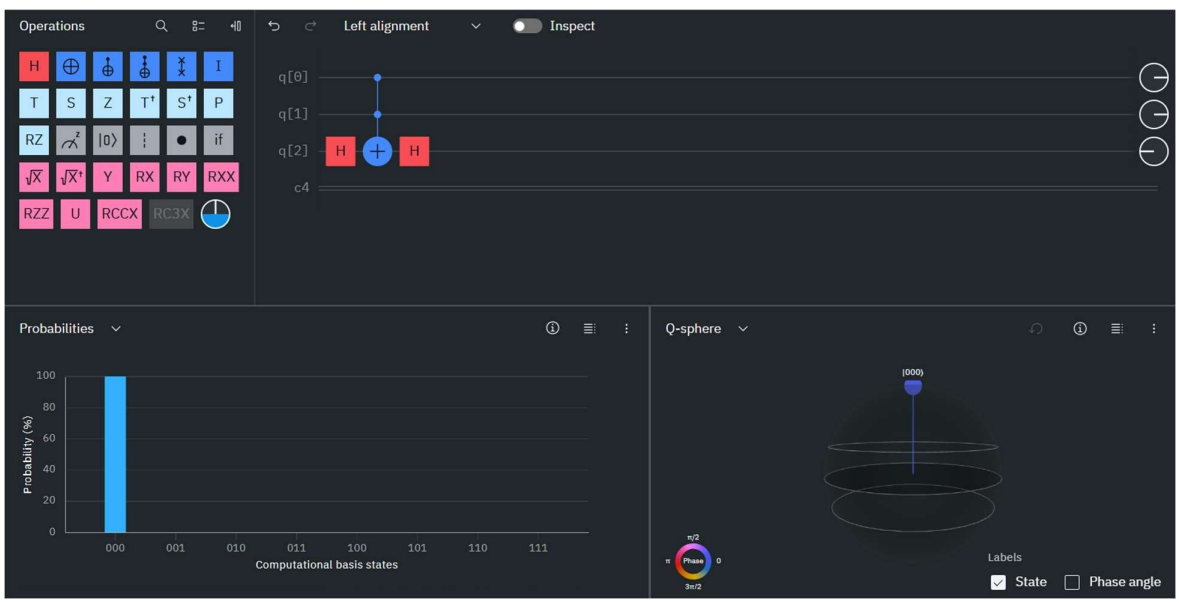
EXPERIMENT 2 :

TOFFOLI GATE: (CCNOT)

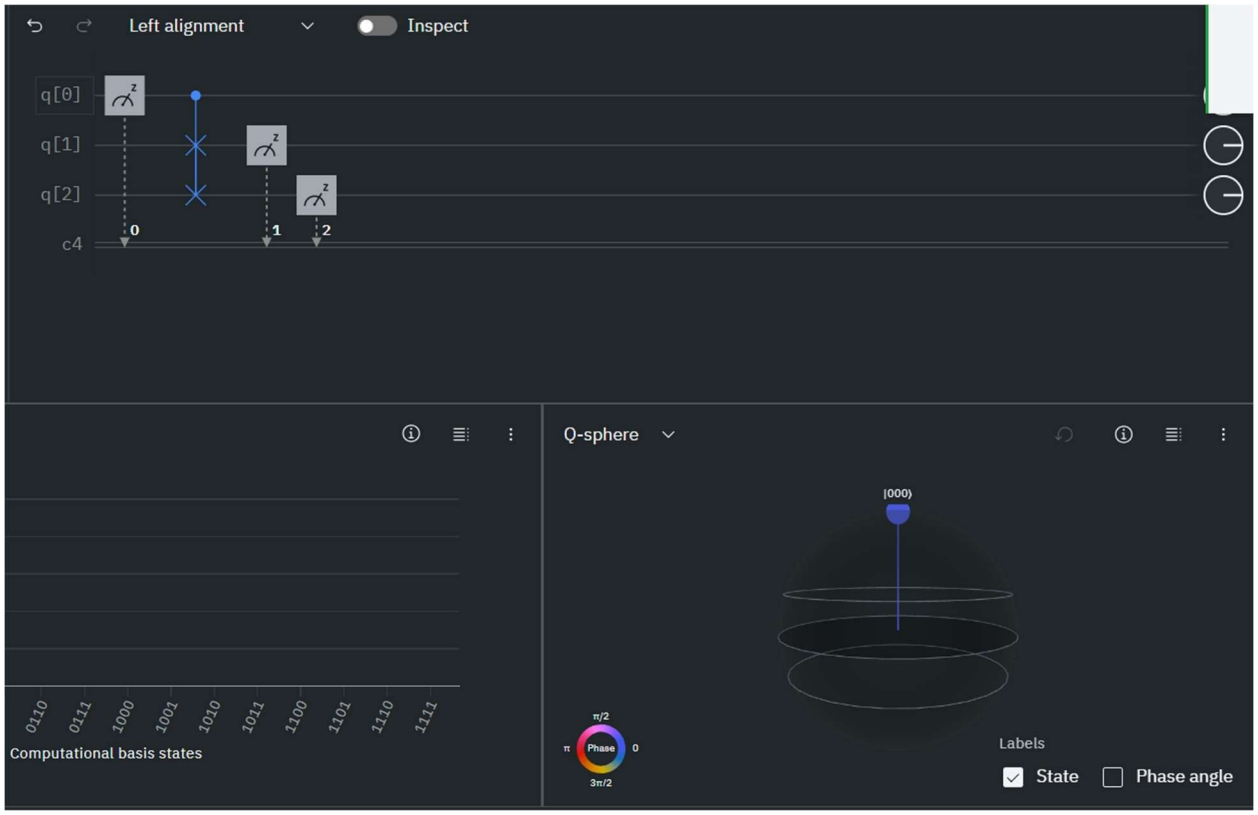



Output histogram for register "c"
Frequency
600
500
400
300
200
100
0
0000
0001
0010
0011
0100
0101
0110
0111
Measurement outcome

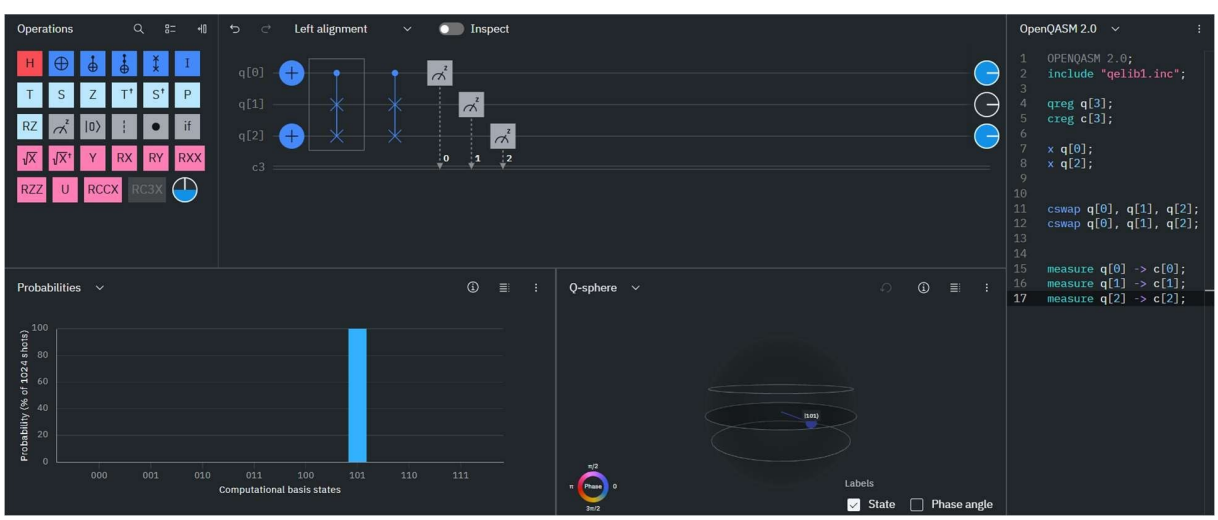
CCZ GATE:


Operations
Left alignment
Inspect
H
I
T
S
Z
T†
S†
P
RZ
|0⟩
if
√X
√X†
Y
RX
RY
RXX
RZZ
U
RCCX
RC3X
q[0]
q[1]
q[2]
c4
Probabilities
Probability (%)
100
80
60
40
20
0
000
001
010
011
100
101
110
111
Computational basis states
Q-sphere
|000⟩
π/2
π
Phase
0
3π/2
Labels
State
Phase angle

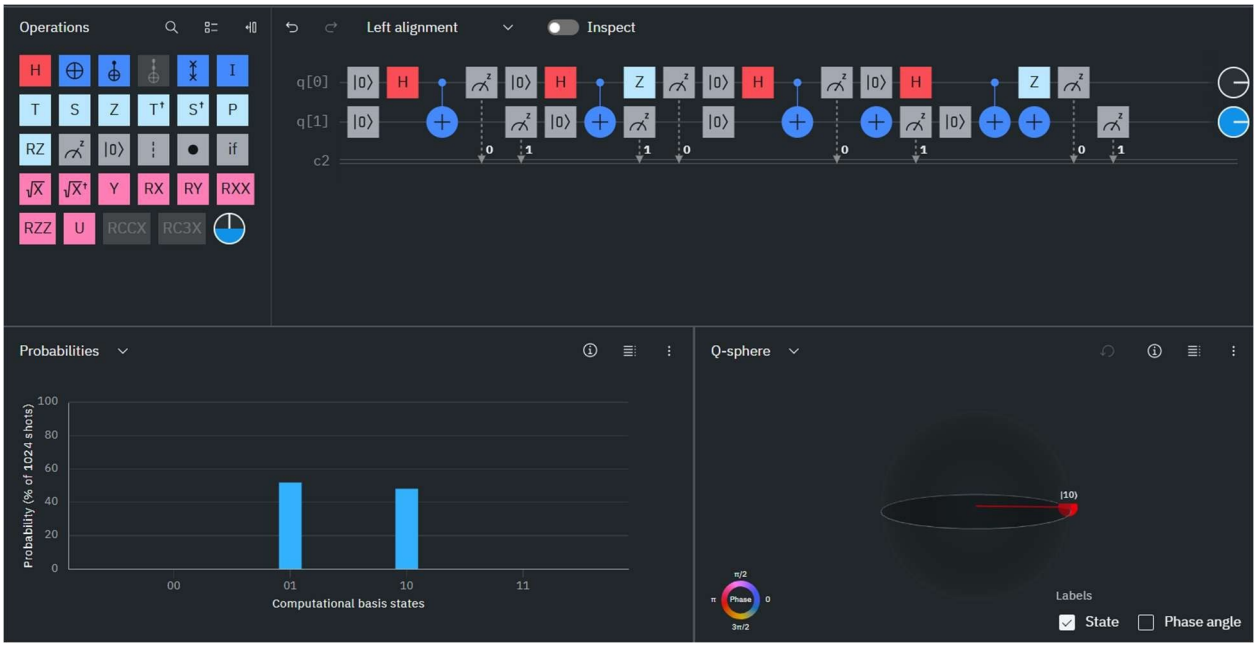
FREDKIN GATE:

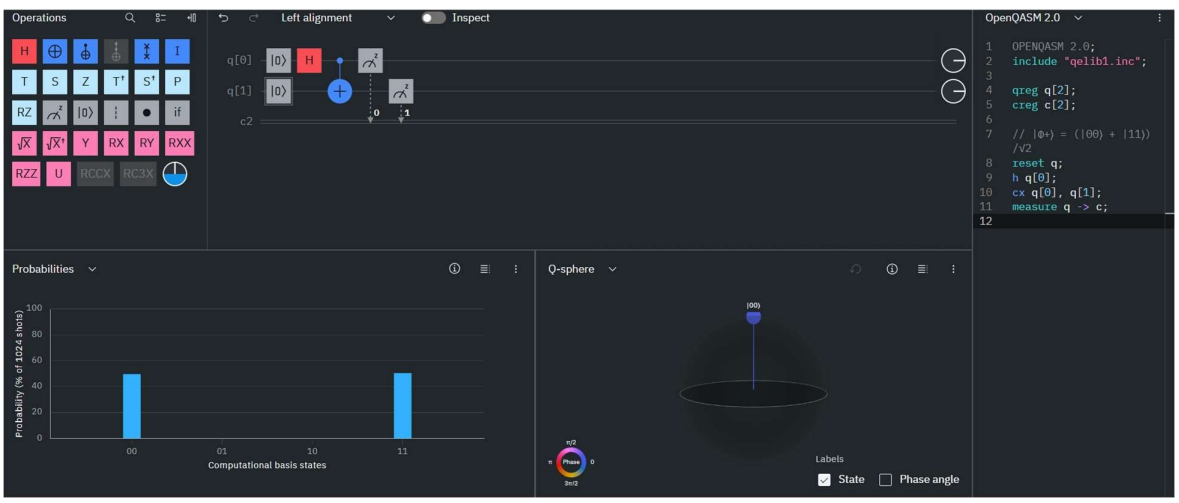


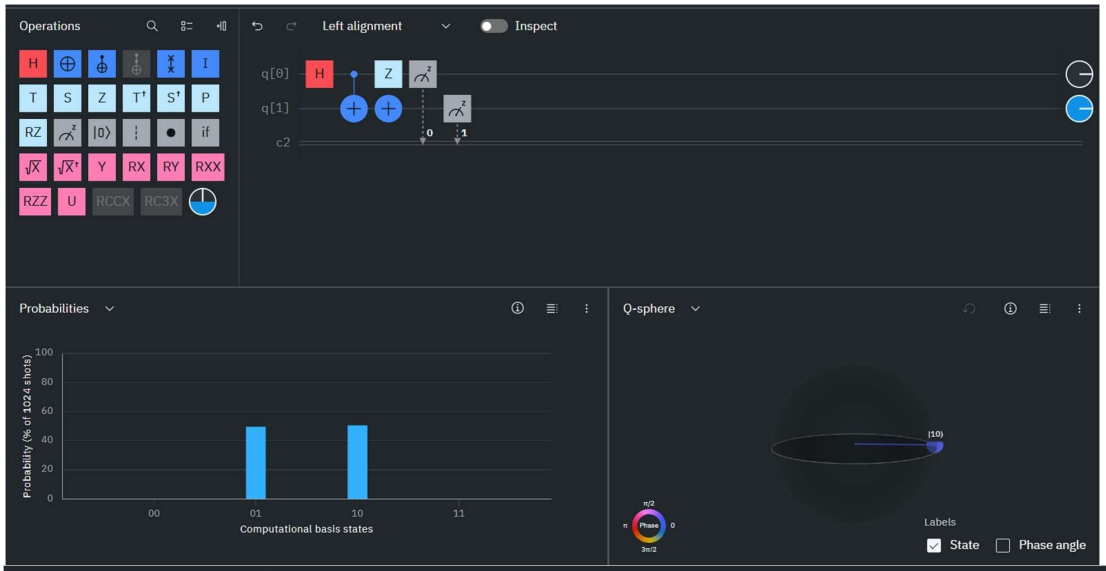
EXPERIMENT 3 :

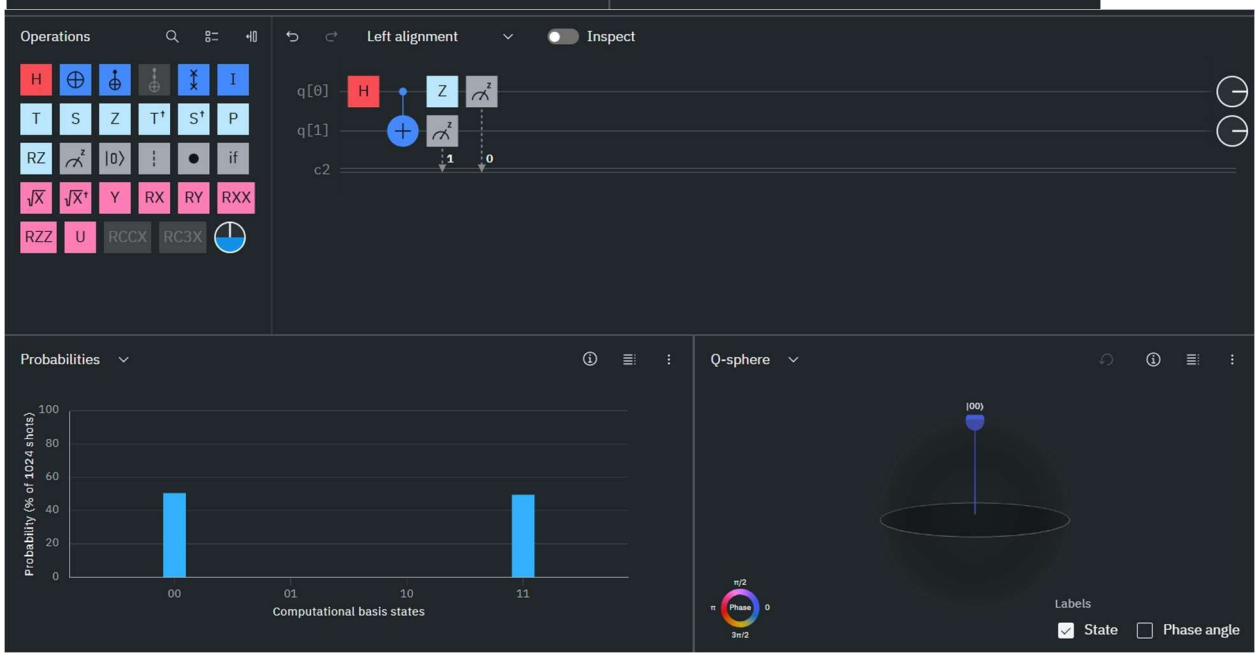


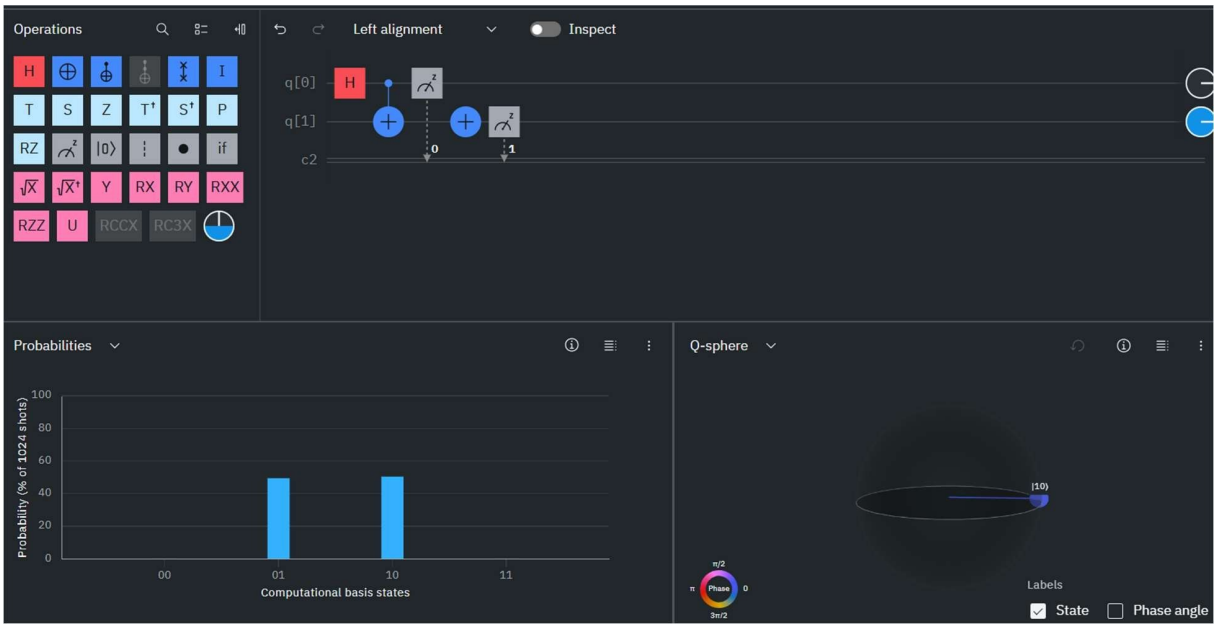
EXPERIMENT 4 :





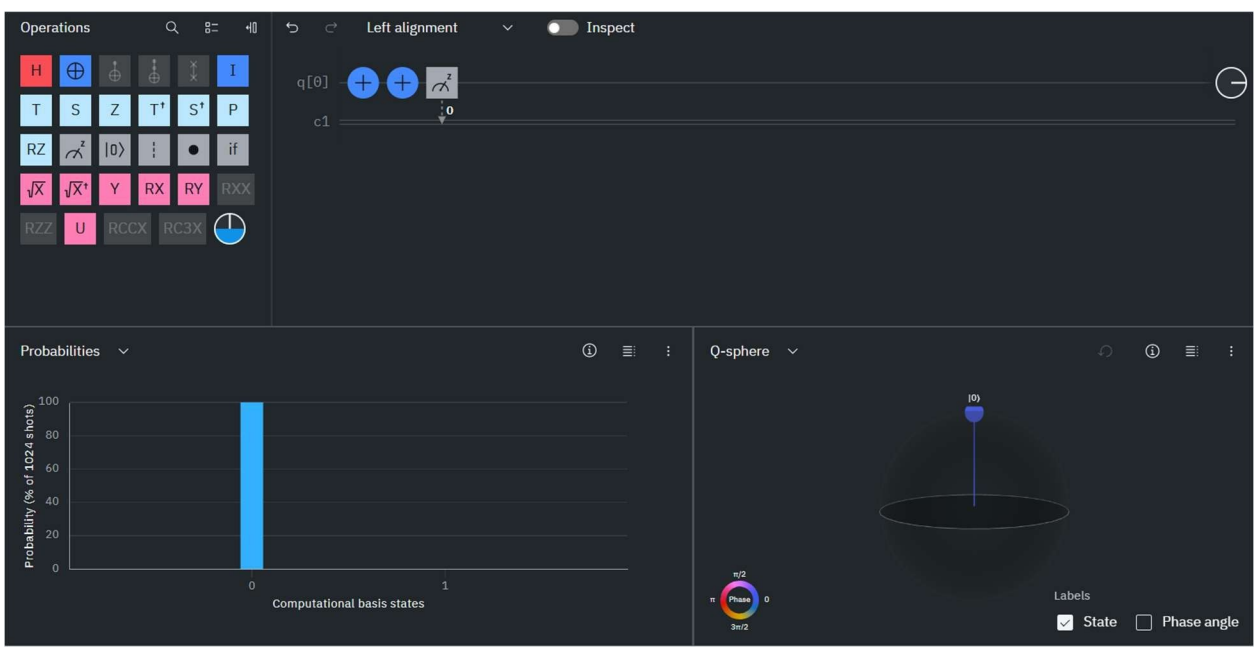

Operations
Left alignment
Inspect
q[0]
q[1]
c2
Probabilities
Probability (% of 1024 shots)
Computational basis states
00
01
10
11
Q-sphere
|10⟩
Labels
State
Phase angle

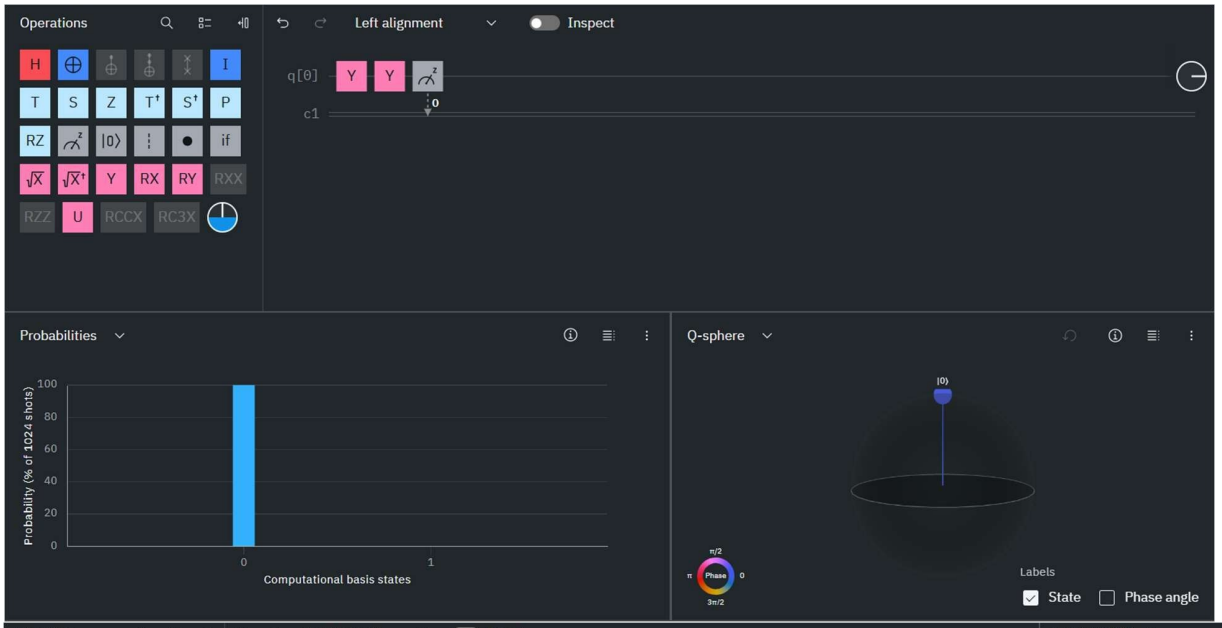

Operations
Left alignment
Inspect
q[0]
q[1]
c2
Probabilities
Probability (% of 1024 shots)
Computational basis states
00
01
10
11
Q-sphere
|00⟩
Labels
State
Phase angle

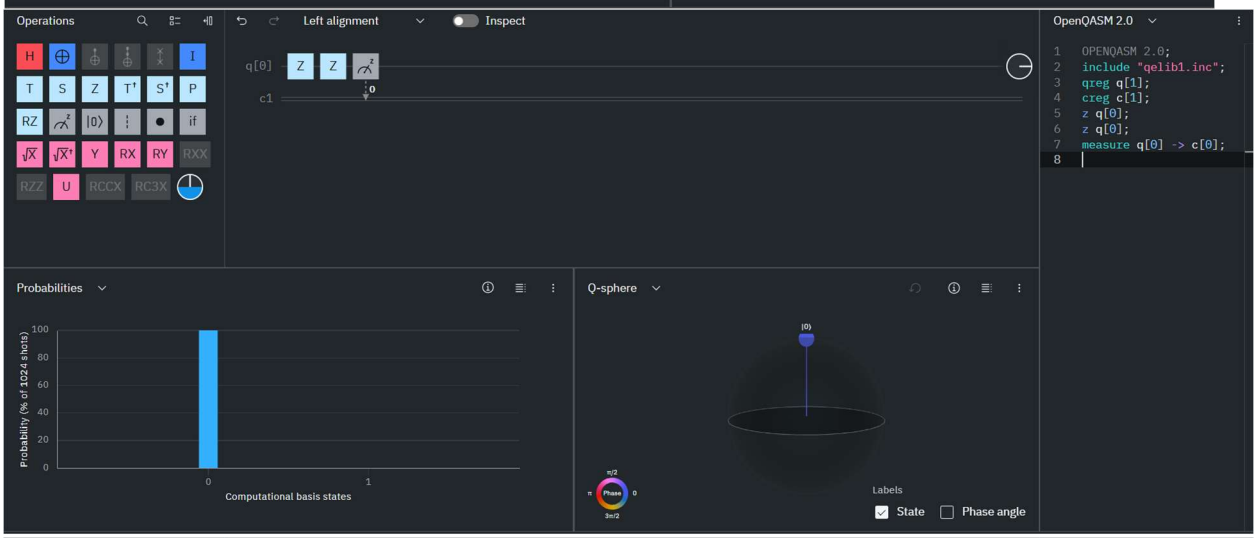

Operations
Left alignment
Inspect
q[0]
q[1]
c2
Probabilities
Probability (% of 1024 shots)
Computational basis states
00
01
10
11
Q-sphere
|10⟩
Labels
State
Phase angle

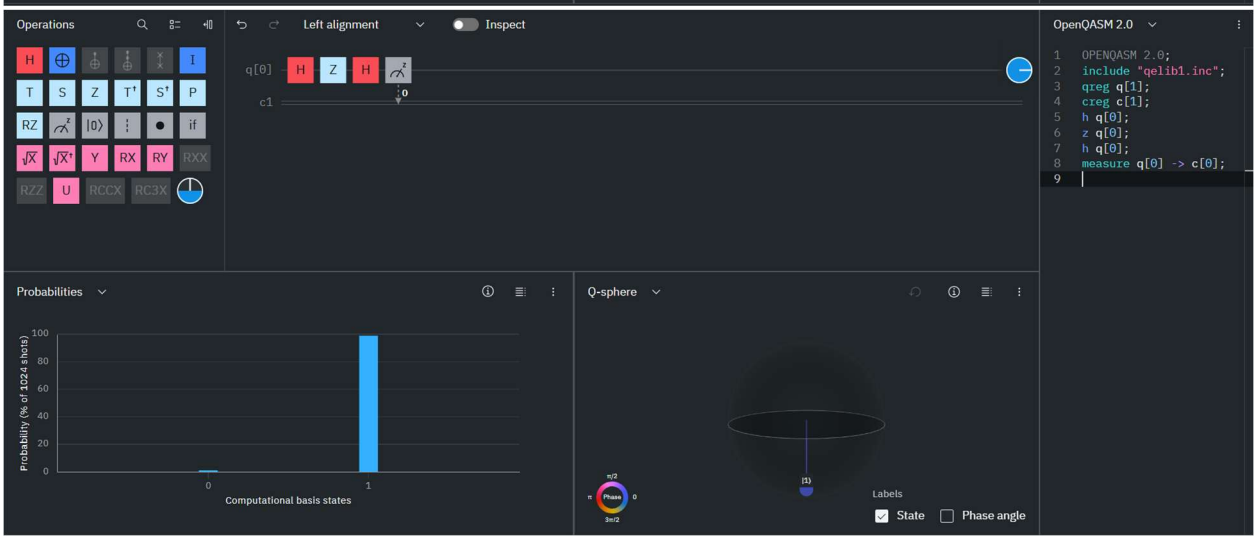
EXPERIMENT 5 :

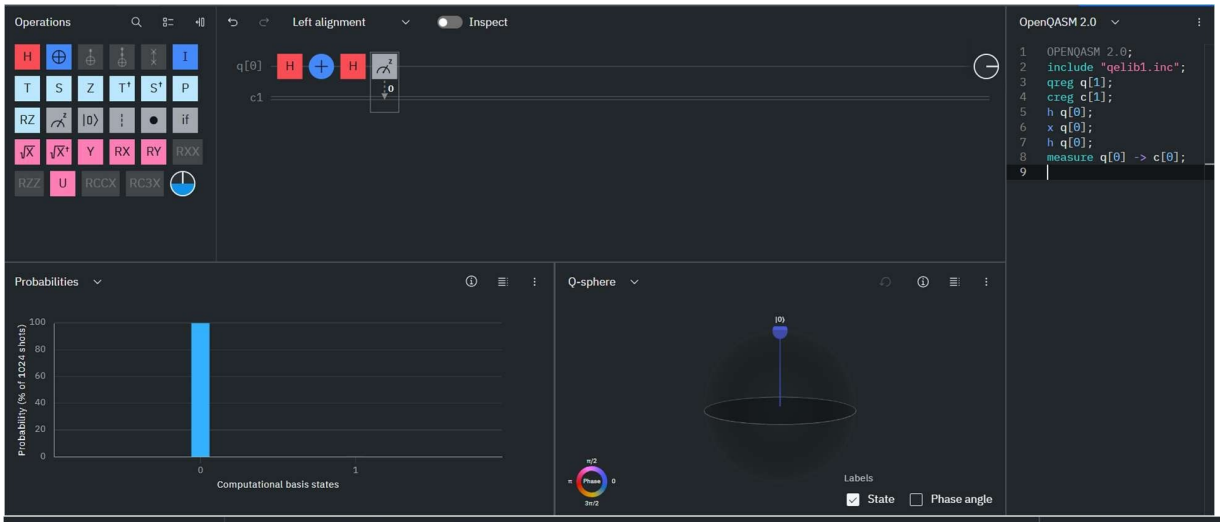
EXPERIMENT 6 :

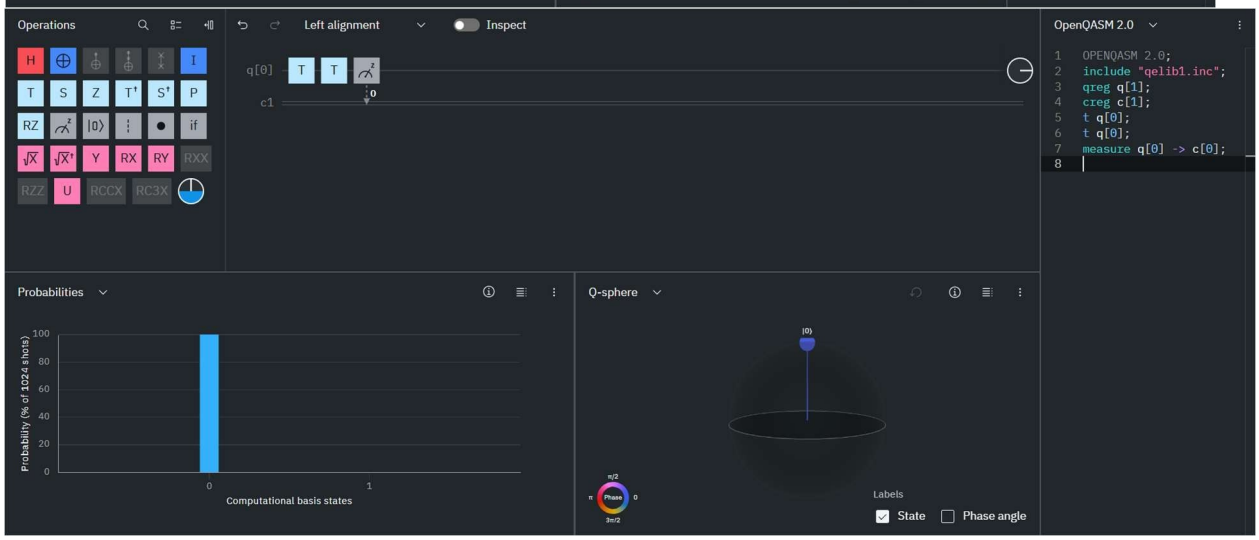


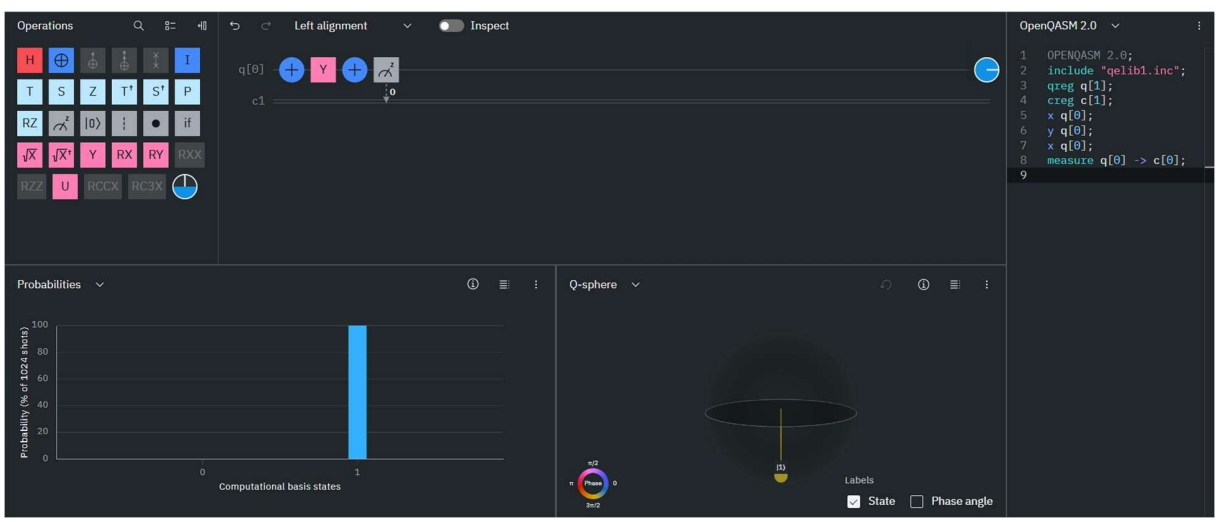
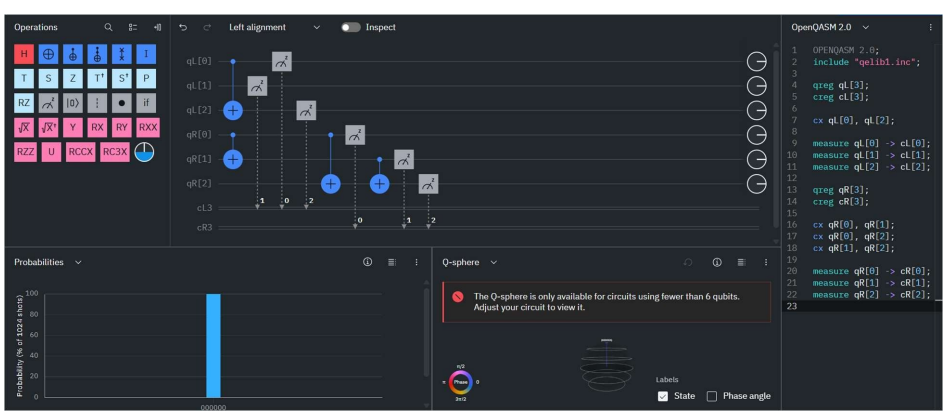

Operations
Left alignment
Inspect
q[0]
c1
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
|0⟩
Labels
State
Phase angle


Operations
Left alignment
Inspect
q[0]
c1
OpenQASM 2.0
OPENQASM 2.0;
include "qelib1.inc";
qreg q[1];
creg c[1];
z q[0];
z q[0];
measure q[0] -> c[0];
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
|0⟩
Labels
State
Phase angle


Operations
Left alignment
Inspect
q[0]
c1
OpenQASM 2.0
OPENQASM 2.0;
include "qelib1.inc";
qreg q[1];
creg c[1];
h q[0];
z q[0];
h q[0];
measure q[0] -> c[0];
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
|1⟩
Labels
State
Phase angle


Operations
Left alignment
Inspect
OpenQASM 2.0
OPENQASM 2.0;
include "qelib1.inc";
qreg q[1];
creg c[1];
h q[0];
x q[0];
h q[0];
measure q[0] -> c[0];
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
|0⟩
Labels
State
Phase angle


Operations
Left alignment
Inspect
OpenQASM 2.0
OPENQASM 2.0;
include "qelib1.inc";
qreg q[1];
creg c[1];
t q[0];
t q[0];
measure q[0] -> c[0];
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
|0⟩
Labels
State
Phase angle


Operations
Left alignment
Inspect
OpenQASM 2.0
OPENQASM 2.0;
include "qelib1.inc";
qreg q[1];
creg c[1];
x q[0];
y q[0];
x q[0];
measure q[0] -> c[0];
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
|1⟩
Labels
State
Phase angle

EXPERIMENT 7 :